Week 5 | **Exercises** p. 247 {6, 8, 16, 40} | **Assessed** Midterm + Final | **File** 4 / 11

4-3

The Multiplication Rules and Conditional Probability

قواعد الضرب والاحتمال الشرطي

§4-2 handled "**or**". §4-3 handles "**and**" — and then inverts it to handle "**given that**". This is the densest section in the chapter and the one that separates a B from an A.

1 · Independent vs dependent events

Independent events

Two events `A` and `B` are **independent** if the fact that A occurs **does not affect** the probability of B occurring.

Dependent events

Two events `A` and `B` are **dependent** if the fact that A occurs **does affect** the probability of B occurring.

IMPORTANT NOTE — INDEPENDENT ≠ MUTUALLY EXCLUSIVE

These two ideas are constantly confused and the lecturer calls it out explicitly. When tossing a **single** coin, the result can be either heads or tails but **cannot be both** — those are *mutually exclusive* events. When tossing **two** coins, the result of one flip **does not affect** the result of the other — those are *independent* events.

The deeper point: mutually exclusive events (with non-zero probabilities) are in fact *dependent*. If A happens, you know for certain B did not — knowing about A changed everything about B. Mutually exclusive is about *overlap*; independent is about *influence*.

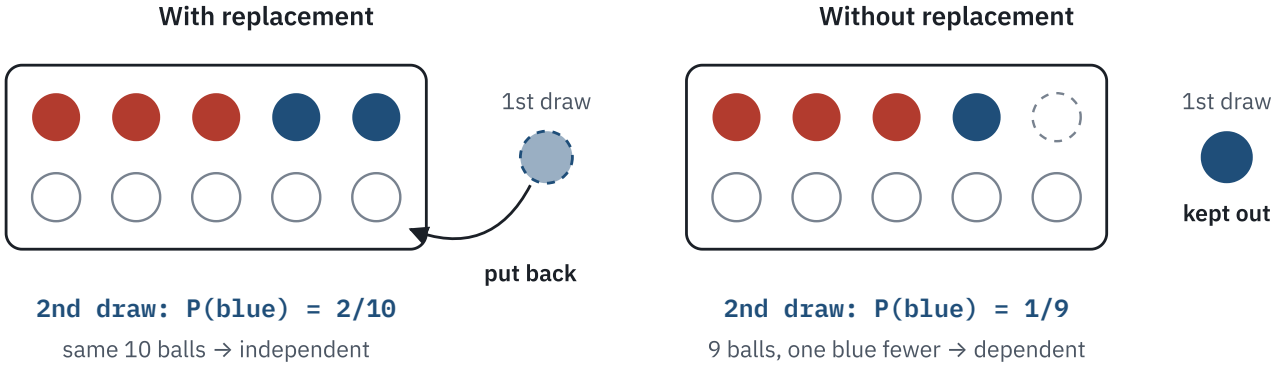
الشرح بالعربي — أهم فكرة في القسم: الاستقلال والتبعية

- **المستقلان (independent):** وقوع أحدهما **لا يغيّر** احتمال الآخر. مثال: رمي عملتين — نتيجة الأولى لا تؤثر في الثانية.
- **التابعان (dependent):** وقوع أحدهما **يغيّر** احتمال الآخر. مثال: سحب كرتين بدون إرجاع — السحبة الأولى تغيّر ما تبقى في الصندوق.

لا تخلط بين المفهومين: «متنافيان» سؤاله «هل يوجد تداخل؟»، و«مستقلان» سؤاله «هل يؤثر أحدهما في الآخر؟». بل إن الحدثين المتنافيين (باحتمالات غير صفرية) يكونان **تابعين**، لأن وقوع A يجعل وقوع B مستحيلًا — أي أنه أثّر فيه بالكامل.

SPOTTING DEPENDENCE IN A WORD PROBLEM

- **"...and then it is replaced"** → independent. The pool resets, so probabilities repeat.
- **"...without replacement"** → dependent. Both the numerator and the denominator shrink on the second draw.
- **Separate devices** (a coin *and* a die, three different people chosen from a large population) → independent.
- **"Of these clients...", "of those who..."** → dependent; you are being given a conditional probability.



Example 4-25's urn (3 red, 2 blue, 5 white) after a blue ball is drawn. Put back, the second draw sees the same urn, so the draws are independent. Kept out, the urn shrinks to 9 with one blue fewer, so the second probability changes — the draws are dependent.

صندوق مثال 25-4 بعد سحب كرة زرقاء. مع الإرجاع: السحبة الثانية من نفس الصندوق (مستقلان). بدون إرجاع: يبقى 9 كرات وزرقاء واحدة فقط، فيتغير الاحتمال (تابعان).

كيف أعرف من صياغة السؤال؟

- **with replacement** / **then it is replaced** (مع الإرجاع) ← **مستقلان**، الاحتمال يتكرر كما هو.
- **without replacement** (بدون إرجاع) ← **تابعان**، البسط والمقام يتغيران في السحبة الثانية.
- أدوات منفصلة (عملة ونرد، أشخاص من مجتمع كبير) ← **مستقلان**.
- **of these** / **of those who** (من بين هؤلاء) ← هذا **احتمال شرطي**، والحدثان تابعان.

2 · The multiplication rules

MULTIPLICATION RULE #1 — INDEPENDENT EVENTS

$$P(A \cap B) = P(A) \cdot P(B)$$

MULTIPLICATION RULE #2 — DEPENDENT EVENTS

$$P(A \cap B) = P(A \mid B) \cdot P(B)$$

$$P(A \cap B) = P(B \mid A) \cdot P(A)$$

Both forms are correct — use whichever conditional probability the question hands you. In practice the question gives one unconditional probability and one conditional one; multiply them.

الشرح بالعربي — قاعدتا الضرب

قاعدة الضرب تجيب عن سؤال: **ما احتمال A و B معًا؟**

- **إذا كانا مستقلين:** $P(A \cap B) = P(A) \times P(B)$. مثال: صورة في العملة و 4 في النرد = $1/2 \times 1/6 = 1/12$.
- **إذا كانا تابعين:** $P(A \cap B) = P(A) \times P(B \mid A)$ ، أي احتمال الأول × احتمال الثاني **بشرط** وقوع الأول. مثال: 53% لديهم تأمين منزل، و**من بين هؤلاء** 27% لديهم تأمين سيارة ← $0.53 \times 0.27 = 0.1431$.

القاعدة الذهبية: and تعني **ضرب**، و **or** تعني **جمع**.

EXAMPLE 4-23 · TOSSING A COIN

A coin is flipped and a die is rolled. Find the probability of getting a head on the coin *and* a 4 on the die.

1. The coin and the die do not influence each other → **independent** → Rule #1.
2. P(head) = 1/2 and P(4) = 1/6.
3. $P(\text{head} \cap 4) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12} \approx$ **0.0833**

Verify by sample space: n(S) = 2 × 6 = 12 outcomes, and only (H, 4) qualifies → 1/12. ✔

EXAMPLE 4-25 · SELECTING A COLORED BALL

An urn contains 3 red balls, 2 blue balls, and 5 white balls. A ball is selected and its color noted. **Then it is replaced.** A second ball is selected and its color noted. Find the probability of: (a) selecting 2 blue balls; (b) selecting 1 blue ball and then 1 white ball; (c) selecting 1 red ball and then 1 blue ball.

Total balls = 3 + 2 + 5 = 10. The ball is *replaced*, so every draw sees the same 10 balls → **independent** → Rule #1.

1. **(a)** $P(B \cap B) = \frac{2}{10} \cdot \frac{2}{10} = \frac{4}{100} = \frac{1}{25} =$ **0.04**
2. **(b)** $P(B \cap W) = \frac{2}{10} \cdot \frac{5}{10} = \frac{10}{100} =$ **0.10**
3. **(c)** $P(R \cap B) = \frac{3}{10} \cdot \frac{2}{10} = \frac{6}{100} =$ **0.06**

THE VARIANT YOU MUST BE READY FOR

If the same question said "**without replacement**", every second denominator becomes 9 and the matching numerator drops by one where the colors agree:
(a) (2/10)(1/9) = 2/90 ≈ 0.0222 · (b) (2/10)(5/9) = 10/90 ≈ 0.1111 · (c) (3/10)(2/9) = 6/90 ≈ 0.0667.
Examiners pair these two versions constantly. Circle the words "replaced" or "without replacement" the moment you see them.

EXAMPLE 4-26 · SURVEY ON STRESS

It was found that 3 out of every 4 American workers say they suffer great stress at least once a week. If three people are selected at random, find the probability that all three will say that they suffer great stress at least once a week.

1. P(stress) = 3/4 = 0.75 for each person; three people drawn from a large population → **independent**.
2. $P(\text{all three}) = \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{27}{64} \approx$ **0.4219**

EXAMPLE 4-27 · MALE COLOR BLINDNESS

Approximately 9% of men have a type of color blindness that prevents them from distinguishing between red and green. If 3 men are selected at random, find the probability that all of them will have this type of red-green color blindness.

1. P(color blind) = 0.09; three independent selections.
2. $P(\text{all three}) = (0.09)^3 = 0.09 \times 0.09 \times 0.09 =$ **0.000729**

Interpretation marks: a result this small should be read aloud in your answer — "about 7 in 10,000 groups of three men". Interpretation is explicitly in CLO8 and costs you nothing to add.

EXAMPLE 4-29 · HOMEOWNER'S AND AUTOMOBILE INSURANCE

World Wide Insurance Company found that 53% of the residents of a city had homeowner's insurance (`H`) with the company. **Of these clients**, 27% also had automobile insurance (`A`). If a resident is selected at random, find the probability that the resident has both homeowner's and automobile insurance with World Wide.

1. **Extract carefully.** P(H) = 0.53. The phrase "*of these clients*" makes the 27% conditional: P(A | H) = 0.27.
2. Dependent → Rule #2.
3. $P(H \cap A) = P(H) \cdot P(A \mid H) = 0.53 \times 0.27 =$ **0.1431**

TRAP — "OF THESE" IS THE CONDITIONAL SIGNAL

If you read the 27% as unconditional P(A) and multiply 0.53 × 0.27 anyway, you get the right number for the wrong reason and will lose the method mark. Worse, on a question where the two routes differ you will get the wrong number. Train yourself to annotate the given values as `P(H) = 0.53` and `P(A|H) = 0.27` *before* calculating.

EXAMPLE 4-28 · UNEMPLOYED WORKERS

For a specific year, 5.2% of U.S. workers were unemployed. During that time, 33% of those who were unemployed received unemployment benefits. If a person is selected at random, find the probability that she or he received unemployment benefits if the person is unemployed.

Extract: P(U) = 0.052, and "33% *of those who were unemployed*" gives P(B | U) = 0.33.

As worded on the slide ("...*if the person is unemployed*"), the question asks for the conditional probability, which is handed to you directly:

$$P(B \mid U) = \boxed{0.33}$$

The textbook version of this example asks for the probability that a randomly selected person "was unemployed **and** received benefits", which is the multiplication rule:

$$P(U \cap B) = P(U) \cdot P(B \mid U) = 0.052 \times 0.33 = \boxed{0.01716 \approx 0.0172}$$

BE READY FOR BOTH

This pair is the clearest illustration in the chapter of how one word changes the whole calculation. "**And**" → multiply the two values (0.0172). "**Given / if / of those who**" → the conditional is already there (0.33). In an exam, write which one you are computing in words before you compute it.

3 · Tree diagrams with branch probabilities

When an experiment happens in stages and the second stage depends on the first, a tree diagram is not optional decoration — it is the method. Multiply *along* branches, add *across* the branches that satisfy the event.

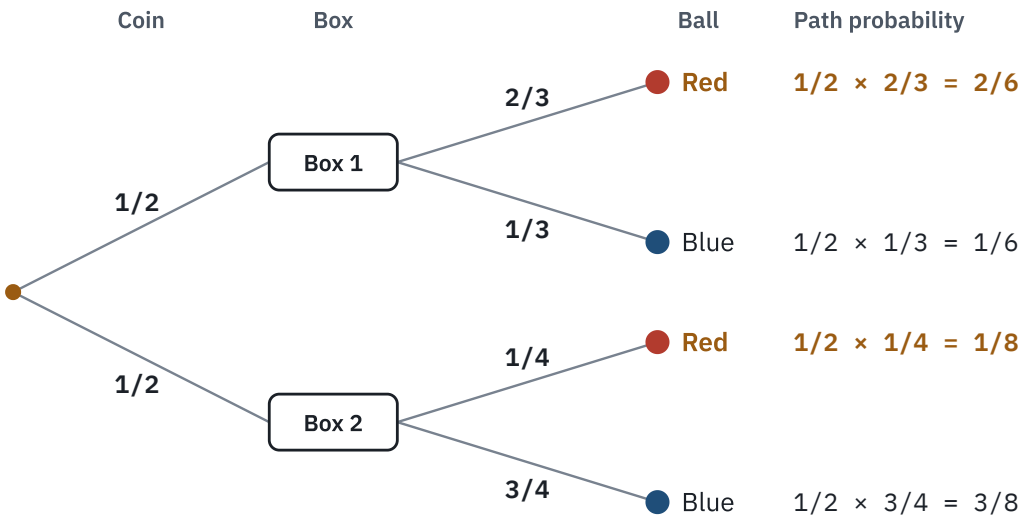
الشرح بالعربي — شجرة الاحتمالات

عندما تتم التجربة على **مراحل** (مثلًا: رمي عملة لاختيار صندوق، ثم سحب كرة منه)، ارسم شجرة:

- **على طول الفرع:** اضرب الاحتمالات — هذا احتمال مسار واحد.
- **بين الفروع:** اجمع احتمالات كل المسارات التي تحقق الحدث المطلوب.
- **للتأكد:** مجموع احتمالات كل المسارات النهائية يجب أن يساوي 1 .

EXAMPLE 4-31 · SELECTING COLORED BALLS

Box 1 contains 2 red balls and 1 blue ball. Box 2 contains 3 blue balls and 1 red ball. A coin is tossed: heads → box 1 is selected and a ball is drawn; tails → box 2 is selected and a ball is drawn. Find the probability of selecting a red ball.



$P(\text{Red}) = 2/6 + 1/8 = 11/24 \approx 0.458$

Multiply along each branch to get a path probability; red is reached by two paths, so add them: 2/6 + 1/8 = 11/24. All four paths add to 1.

اضرب على طول كل فرع لتحصل على احتمال المسار. الكرة الحمراء لها مساران، فنجمعهما: 2/6 + 1/8 = 11/24. ومجموع المسارات الأربعة = 1.

Red can be reached by two different paths, so add them:

$$P(\text{Red}) = \frac{1}{2}\cdot\frac{2}{3} + \frac{1}{2}\cdot\frac{1}{4} = \frac{2}{6} + \frac{1}{8} = \frac{8}{24} + \frac{3}{24} = \frac{11}{24} \approx \boxed{0.4583}$$

Check: all four products sum to 1/3 + 1/6 + 1/8 + 3/8 = 8/24 + 4/24 + 3/24 + 9/24 = 24/24 = 1. ✔ A tree whose branches don't sum to 1 has an error in it.

4 · Conditional probability

The **conditional probability** of an event `B` in relationship to an event `A` is the probability that event B occurs *after* event A has already occurred.

CONDITIONAL PROBABILITY

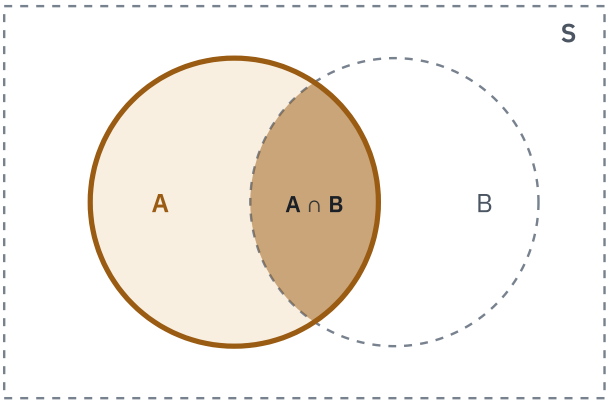
$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} \qquad P(A) \neq 0$$

Read it as: *the condition goes in the denominator.* Conditioning on A means you throw away everything outside A and treat A as the new sample space.

CONDITIONAL PROBABILITY AND INDEPENDENCE

If A and B are independent: $P(B \mid A) = \frac{P(A \cap B)}{P(A)} = P(B)$

Knowing A happened tells you nothing new about B — which is exactly what "independent" means. This doubles as the **test for independence**.



Once A has happened, A is the new **sample space**

P(B | A) = P(A ∩ B) ÷ P(A)

Conditioning on A throws away everything outside A. P(B | A) is the share of circle A that is also in B: the shaded lens divided by the whole of A.

عند الشرط «بشرط A» نتجاهل كل ما هو خارج A، وتصبح A فضاء العينة الجديد. P(B|A) = المنطقة المشتركة ÷ كامل A.

الشرح بالعربي — الاحتمال الشرطي

`P(B | A)` يُقرأ «احتمال B **بشرط** أن A قد وقع». معناه أننا نعرف مسبقًا أن A حدث، فنحصر أنفسنا داخل A فقط ونعتبره فضاء العينة الجديد.

الشرط دائمًا في المقام — `P(B | A) = P(A ∩ B) ÷ P(A)`

مثال: احتمال أن يوقف سام سيارته في مكان ممنوع **و** يحصل على مخالفة `0.06` ، واحتمال أن يوقفها في مكان ممنوع `0.20` . إذا أوقفها فعلًا في مكان ممنوع، فاحتمال المخالفة = `0.30 = 0.20 ÷ 0.06` .

اختبار الاستقلال: إذا كان `P(B | A) = P(B)` — أو `P(A ∩ B) = P(A) × P(B)` — فالحدثان مستقلان، أي أن معرفة A لم تغيّر شيئًا.

كلمات الشرط في السؤال: given that، if، of those who، أو حقيقة مذكورة على أنها وقعت («وصل سام واضطر للوقوف في مكان ممنوع»).

EXAMPLE 4-32 · SELECTING COLORED CHIPS

A box contains black chips and white chips. A person selects two chips **without replacement**. If the probability of selecting a black chip and a white chip is 15/56, and the probability of selecting a black chip on the first draw is 3/8, find the probability of selecting the white chip on the second draw, *given* that the first chip selected was a black chip.

1. **Extract:** P(B ∩ W) = 15/56, P(B) = 3/8. Required: P(W | B).
2. $$P(W \mid B) = \frac{P(B \cap W)}{P(B)} = \frac{15/56}{3/8} = \frac{15}{56} \times \frac{8}{3} = \frac{120}{168} = \frac{5}{7} \approx \boxed{0.714}$$

Dividing by a fraction: flip the bottom fraction and multiply. 15/56 ÷ 3/8 = 15/56 × 8/3. Cancel: 15/3 = 5 and 8/56 = 1/7 → 5/7.

EXAMPLE 4-33 · PARKING TICKETS

The probability that Sam parks in a no-parking zone *and* gets a parking ticket is 0.06, and the probability that Sam cannot find a legal parking space and has to park in the no-parking zone is 0.20. On Tuesday, Sam arrives at school and has to park in a no-parking zone. Find the probability that he will get a parking ticket.

1. **Extract:** Let N = parks in a no-parking zone, T = gets a ticket. P(N ∩ T) = 0.06, P(N) = 0.20.
2. "He *has to* park in a no-parking zone" = N has already happened → the question wants P(T | N).
3. $$P(T \mid N) = \frac{P(N \cap T)}{P(N)} = \frac{0.06}{0.20} = \boxed{0.30}$$

Why this is the exam's favourite shape: the condition is hidden inside a story ("he has to park there") rather than announced with the word "given". Learn to spot a condition stated as an accomplished fact.

5 · Two-way (contingency) tables — the complete toolkit

A two-way table lets you answer every question type in §4-1 to §4-3 by reading counts.
Master the four readings and this becomes free marks.

EXAMPLE 4-34 · SURVEY ON WOMEN IN THE MILITARY

A recent survey asked 100 people if they thought women in the armed forces should be permitted to participate in combat:

GENDER	YES	NO	TOTAL
Male	32	18	50
Female	8	42	50
Total	**40**	**60**	**100**

Find: (a) the respondent answered yes, given that the respondent was female; (b) the respondent was male, given that the respondent answered no; (c) are the events "Male" and "Yes" independent? (d) are "Male" and "Yes" mutually exclusive?

1. **(a) P(Yes | Female).** The condition "Female" restricts you to the Female row, total 50. Of those, 8 said yes.

   $$P(\text{Yes} \mid \text{Female}) = \frac{8}{50} = \boxed{0.16}$$

2. **(b) P(Male | No).** The condition "No" restricts you to the No column, total 60. Of those, 18 are male.

   $$P(\text{Male} \mid \text{No}) = \frac{18}{60} = \boxed{0.30}$$

3. **(c) Independent?** Test whether P(M ∩ Y) = P(M) · P(Y).

   $$P(M \cap Y) = 32/100 = 0.32$$

   $$P(M) \cdot P(Y) = (50/100)(40/100) = 0.5 \times 0.4 = 0.20$$

   $0.32 \neq 0.20 \rightarrow$ **Not independent — the events are dependent.**

   Equivalent test: P(Y | M) = 32/50 = 0.64 but P(Y) = 40/100 = 0.40. Knowing the respondent is male changes the probability of "yes", so they are dependent.

4. **(d) Mutually exclusive?** P(M ∩ Y) = 32/100 = 0.32 ≠ 0. There are 32 people who are *both* male and said yes.

   → **Not mutually exclusive.**

PARTS (C) AND (D) TOGETHER ARE THE WHOLE POINT OF THE EXAMPLE

One table, two different questions that students merge into one. **Mutually exclusive** asks "*is the overlap empty?*" — look at the cell. **Independent** asks "*does one change the other?*" — compare a conditional with its unconditional version, or compare P(A∩B) with P(A)P(B). Answering (d) with the independence test, or (c) with the overlap test, scores zero.

THE FOUR READINGS OF A TWO-WAY TABLE

QUESTION TYPE	WHAT GOES IN THE DENOMINATOR	EXAMPLE FROM 4-34
Simple P(A)	The grand total.	P(Male) = 50/100
Joint "and" P(A ∩ B)	The grand total; numerator is the single cell.	P(Male ∩ Yes) = 32/100
Union "or" P(A ∪ B)	The grand total; add the row and column, subtract the shared cell once.	P(Male ∪ Yes) = (50 + 40 − 32)/100 = 0.58
Conditional "given" P(A | B)	The *row or column total* of the condition.	P(Yes | Female) = 8/50

القاعدة الذهبية في جداول التقاطع

الشرط يحدد المقام. إذا قيل "given that it is a female" فالمقام هو مجموع صف الإناث (50) وليس المجموع الكلي (100). أما "and" فالمقام يبقى المجموع الكلي والبسط هو الخلية المشتركة.

و**فرّق دائمًا** بين: **mutually exclusive** (هل التقاطع فارغ؟ انظر إلى الخلية) و**independent** (هل يؤثر أحدهما في الآخر؟ قارن $P(A \cap B)$ مع $P(A) \cdot P(B)$).

6 · Probabilities for "at least" — the complement shortcut

This is the technique the lecture notes flag under "Other Examples: Probabilities for *At Least*" (textbook Examples 4-36 *Rolling a Die* and 4-37 *Ties*, set for self-study). It is the single most reusable trick in the chapter.

THE "AT LEAST ONE" RULE

$$P(\text{at least one}) = 1 - P(\text{none})$$

"At least one" has many ways to happen (exactly one, exactly two, exactly three, ...) but its complement — "none" — has exactly one. Compute the easy side and subtract.

الشرح بالعربي — «واحد على الأقل»

«واحد على الأقل» له طرق كثيرة للحدوث (واحد، اثنان، ثلاثة...)، لكن عكسه «ولا واحد» له طريقة **واحدة فقط**. لذلك:

$$P(\text{at least one}) = 1 - P(\text{none})$$

مثال: رمي نرد 3 مرات، احتمال ظهور 5 مرة واحدة على الأقل $= 1 - (5/6)^3 = 1 - 125/216 \approx 0.4213$.

كلما رأيت **at least** في السؤال، فكّر في المتمّمة قبل أي شيء آخر.

ILLUSTRATION · THE STANDARD SHAPE

A single die is rolled 3 times. Find the probability of getting **at least one** 5.

1. **The long way (don't):** P(exactly one 5) + P(exactly two 5s) + P(three 5s). Three separate calculations.
2. **The complement way:** "at least one 5" has complement "no 5s at all". P(not a 5 on one roll) = 5/6.
3. $P(\text{no 5s in 3 rolls}) = (5/6)^3 = \frac{125}{216}$
4. $P(\text{at least one 5}) = 1 - \frac{125}{216} = \frac{91}{216} \approx \mathbf{0.4213}$



P(at least one 5) = 1 − 125/216 = 91/216 ≈ 0.421

Drawn to scale for three rolls of a die. “None” is a single case you can compute in one step; “at least one” is everything else, so it equals 1 − P(none).

مرسوم بالنسبة الحقيقية لثلاث رميات: «ولا مرة» حالة واحدة سهلة الحساب، و«مرة على الأقل» هي كل الباقي، أي 1 − P(none).

The same pattern solves "at least one defective", "at least one female", "at least one match" — see Example 4-51(d) in §4-5, where it appears in its examined form.

THE THREE COMPLEMENT TRIGGERS

Reach for `1 − P(…)` the moment you read: **"at least one"** (complement = none) · **"not"** (complement = the event itself) · **"neither / none"** (complement = at least one). Scanning for these three phrases before you start writing will save you minutes on every paper.

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests

01 · Lesson 4-1 — Sample Spaces and Probability

02 · Lesson 4-2 — The Addition Rules

03 · Lesson 4-3 — Multiplication Rules and Conditional Probability ← this file

04 · Lesson 4-4 — Counting Rules

05 · Lesson 4-5 — Probability and Counting Rules

06 · Master formula sheet

07 · Which rule do I use?

08 · The 14 traps that cost marks

09 · Review exercises, solved

10 · The A+ study protocol

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.